V Tutorial

Weak stationarity, autocorrelations and autoregressive forecasting

Exercise 1. Weak stationarity

Let ε_t be i.i.d. over integer time, with $E[\varepsilon_t] = 0$, $\mathrm{Var}(\varepsilon_t) = \sigma^2 > 0$ and $E[\varepsilon_t^4] < \infty$. The parameters a, b, θ, μ are fixed constants. For each process below, derive its mean, variance and autocovariance $\mathrm{Cov}(y_t, y_{t-h})$ for admissible $h \geq 0$. Determine whether it is weakly stationary, stating any exceptional parameter values.

(a) $y_t = \varepsilon_t^2$.

(b) $y_t = t\varepsilon_t$.

(c) $y_t = a + bt + \varepsilon_t$.

(d) $y_t = a\,\mathbf{1}\{t > 50\} + \varepsilon_t$.

(e) $y_t = \varepsilon_t + \theta\varepsilon_{t-1}$, an MA(1) process.

(f) $y_t = \mu + \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}$, $|\phi| < 1$: the stationary solution of $y_t - \mu = \phi(y_{t-1} - \mu) + \varepsilon_t$.

(g) $y_t = y_{t-1} + \varepsilon_t$, $t \geq 1$, with $y_0 = 0$.

(h) $y_t = \varepsilon_t + (-1)^t \varepsilon_{t-1}$.

In particular, explain why serial dependence does not rule out weak stationarity, and why constant mean and variance alone do not establish it.

Exercise 2. Autocorrelations of daily market returns

Notebook: Autocorrelations.ipynb

Analyse daily US market log returns for 2021–2025 (1,255 trading sessions). Inspect the return series and its ACF at lags 1–15. Use the five-lag Ljung–Box test to assess evidence against jointly zero autocorrelations, and compare the ACF of returns with that of squared demeaned returns. Discuss the interpretation and limitations of the observed evidence.

Relevant concepts

- Simple and log returns; total and excess returns; trading-session lags.
- Population and sample autocorrelations; ACF; pointwise confidence bands.
- Joint hypotheses; Ljung–Box statistic; reference distribution; p-values.
- Linear dependence versus independence; squared returns as a variance proxy.
- Volatility clustering; test assumptions; limits of market-efficiency conclusions.

Exercise 3. US real GNP growth dynamics and forecasting

Notebook: GNP_Growth_AR_Forecasting.ipynb

Analyse quarterly US real GNP log growth for 1947Q2–2019Q4 (291 quarters). Compare the BIC-selected AR(1) and AIC-selected AR(3) on a common estimation sample. Interpret their coefficients, persistence and next-quarter forecasts, then examine sensitivity to extending the sample through 2025Q4.

Relevant concepts

- GNP versus GDP; levels versus growth; quarterly versus annualised growth.
- ACF and PACF; autoregressive models; conditional OLS; common estimation sample.
- AR(p); AIC and BIC; lag selection versus residual adequacy.
- Intercept and lag coefficients; partial associations; long-run mean; uncertainty.
- Stability; persistence; impulse responses.
- Forecast origin; one-step forecasts; held-out outcomes; forecast errors.
- Prediction versus coefficient confidence intervals; parameter and order uncertainty.
- Revised data versus real-time forecasting; sample sensitivity; exceptional events.

Optional extension. The oscillatory period implied by AR(3)

Notebook: GNP_Growth_AR_Forecasting.ipynb

Examine the oscillatory component of the fitted AR(3). Compare its implied period with the speed of damping and the enlarged impulse-response tail, distinguishing a model-implied oscillation from observed business-cycle duration.

Relevant concepts

- Dynamic eigenvalues; complex roots; oscillatory period.
- Amplitude-envelope half-life; damping; model-implied versus observed cycles.

Solutions

Exercise 1. Weak stationarity

Weak stationarity requires finite second moments, constant mean and variance, and autocovariance depending only on the lag. Below, $C_t(h) = \mathrm{Cov}(y_t, y_{t-h})$, with $h \geq 0$.

(a) Squared i.i.d. innovations

$E[y_t] = \sigma^2, \qquad \mathrm{Var}(y_t) = E[\varepsilon_t^4] - \sigma^4, \qquad C_t(h) = 0 \quad (h \geq 1).$

Distinct squared innovations are independent. The finite fourth moment ensures finite variance, so the process is **weakly stationary**.

(b) Time-dependent scale

$E[y_t] = 0, \qquad \mathrm{Var}(y_t) = t^2\sigma^2, \qquad C_t(h) = 0 \quad (h \geq 1).$

The variance changes with t: **not weakly stationary**, despite zero mean and zero autocovariances at positive lags.

(c) Deterministic linear trend

$E[y_t] = a + bt, \qquad \mathrm{Var}(y_t) = \sigma^2, \qquad C_t(h) = 0 \quad (h \geq 1).$

Not weakly stationary if $b \neq 0$, because the mean changes with time; weakly stationary if $b = 0$.

(d) A break in the mean

$E[y_t] = a\,\mathbf{1}\{t > 50\}, \qquad \mathrm{Var}(y_t) = \sigma^2, \qquad C_t(h) = 0 \quad (h \geq 1).$

Not weakly stationary if $a \neq 0$ over a time domain spanning the mean break; weakly stationary if $a = 0$.

(e) MA(1): stationary serial dependence

The mean is zero. Independence gives $\gamma(0) = (1 + \theta^2)\sigma^2, \qquad \gamma(1) = \theta\sigma^2, \qquad \gamma(h) = 0 \quad (h \geq 2).$

Only ε_{t-1} is shared at lag one. The process is **weakly stationary for every finite** θ; serial dependence is allowed.

(f) AR(1): the stationary solution

$E[y_t] = \mu, \qquad \gamma(0) = \sigma^2 \sum_{j=0}^{\infty} \phi^{2j} = \dfrac{\sigma^2}{1 - \phi^2}, \qquad \gamma(h) = \phi^h \gamma(0).$

Since $|\phi| < 1$, the variance sum is finite and moments do not depend on t: **weakly stationary**. The stated infinite-past construction avoids the initial transition of an AR(1) started at a fixed value.

(g) Random walk with a fixed initial value

From $y_0 = 0$, $y_t = \sum_{j=1}^{t} \varepsilon_j$, so $E[y_t] = 0, \qquad \mathrm{Var}(y_t) = t\sigma^2, \qquad C_t(h) = (t - h)\sigma^2 \quad (0 \leq h \leq t).$

The shared $t - h$ innovations give the covariance. Variance and autocovariance depend on t: **not weakly stationary**. Its first difference is ε_t, which is weakly stationary.

(h) Constant mean and variance, changing autocovariance

$E[y_t] = 0, \qquad \mathrm{Var}(y_t) = 2\sigma^2, \qquad C_t(1) = (-1)^t\sigma^2.$

Only ε_{t-1} is shared at lag one; $C_t(h) = 0$ for $h \geq 2$. The lag-one covariance alternates with t: **not weakly stationary**, despite constant mean and variance.